

Santiago HERRERA

Software Engineer | Full Stack, Systems Design, Distributed Systems

[in linkedin.com/in/santipaulh](https://www.linkedin.com/in/santipaulh)

+593 99 551 9509

<https://santipaul.com/en>

[@santihm2006@gmail.com](mailto:santihm2006@gmail.com)

Ecuador, Guayas



Self-taught Full Stack Software Engineer and Software Engineering student with 6 years of experience in Node.js, TypeScript, and automation systems.

I have been building backend systems, automation, and scalable tools since I was 13–17 years old, including bots and infrastructure for high-traffic Discord communities, API integrations, and my own SaaS platform for automation and CRM.

WORK EXPERIENCE

Feb 2023
Apr 2023

Technical Support & Infrastructure, LICEO LOS DELFINES, Ecuador

- **Role purpose:** IT infrastructure support and administration of educational systems, ensuring operational continuity of network services, academic platforms, and workstations.

Key achievements:

- Identified and fixed a critical vulnerability in the academic system (Gescolar) that allowed unauthorized access via authentication tokens.
- Implemented **Windows Server 2022 and Active Directory** for centralized user and device management.
- Established security policies across campus services and file systems.
- Deployed and standardized workstation infrastructure across classrooms and administrative offices.
- Implemented LAN monitoring using **Veyon** for real-time supervision of devices.

Windows Server Active Directory IT Support Networking TCP/IP Cybersecurity Systems Administration

Feb 2024
Aug 2024

Freelance Automation & CRM Systems Developer, MULTIPLE CLIENTS, Ecuador

- **Role purpose:** Design and implementation of marketing automation, CRM, and client management systems to optimize lead capture, tracking, and conversion.

Key achievements:

- Built end-to-end lead automation flows (capture → follow-up → conversion → scheduling), reducing manual operational work.
- Implemented chatbots and conversational flows for automated customer support.
- Integrated **Google Calendar API and Zapier**, eliminating manual scheduling processes.
- Designed acquisition funnels using Meta Ads optimized for conversion.
- Worked with clients in **healthcare and service industries**, adapting solutions to each business workflow (Veterinaria Las Lomas, Odontología Julio Moncayo, Guardería Danny Kids).

CRM Automation APIs Google Calendar API Zapier Botpress GoHighLevel System Integration
Workflow Automation

PROJECTS & INFORMAL EXPERIENCE

May 2019
Feb 2023

Software Engineering (Discord Systems), SELF-DIRECTED / INDEPENDENT PROJECTS, Remote

- **Role purpose:** Design, scaling, and automation of high-traffic Discord communities for content creators, focused on moderation, stability, and engagement at scale.

Key achievements:

- Scaled Discord communities from **4k/10k to 168k/223k users** through moderation and automation systems.
- Developed bots in **Node.js and TypeScript** for moderation, security, and community workflows since early stages (age 13).
- Managed systems with **+20k concurrent users**, ensuring stability during traffic spikes.
- Implemented **rate limiting, event queues, and sharding**, reducing failures under high load.
- Built VPS infrastructure (Hetzner / OVH) using **Docker and Linux**, improving deployment and debugging workflows.

Node.js TypeScript Discord API System Design Sharding Rate Limiting Redis Docker Linux
VPS Management

2024
Present

Full Stack SaaS Platform (Automation & CRM Infrastructure), INDEPENDENT PROJECT, Remote

> **Role purpose:** Built a unified SaaS platform for automation and CRM to replace external tools and centralize business workflows and lead management.

Key achievements:

> Migrated automation from external tools (Zapier, Botpress, Google Calendar API) into a unified in-house architecture.

> Designed a scalable full-stack system using **Node.js, TypeScript, and Next.js**, improving modularity and maintainability.

> Built backend with **PostgreSQL (Supabase)** and **Redis** for real-time data, caching, and structured persistence.

> Developed frontend in Next.js with **Tailwind CSS** and Zustand/Redux for state management and UI performance.

> Containerized the system with **Docker**, ensuring consistent environments across development and deployment.

> Designed event-driven architecture and internal APIs for end-to-end workflow automation.

Node.jsTypeScriptNext.jsPostgreSQL (Supabase)RedisDockerAPI DesignSystem Architecture

Full Stack DevelopmentEvent-Driven Architecture

 LANGUAGES

English

Spanish

B2

Native

 TECH SKILLS

Node.js

TypeScript

React.js / Next.js

HTML/CSS/JavaScript

API Integration (CRUD)

PostgreSQL / MongoDB / MySQL

Python

PyTorch

Java

PHP / Laravel

CRM Automation

GoHighLevel)

Networking

REST APIs, Authentication, Backend Systems, Express

Scalable Applications, Backend, Frontend Development

Frontend Development, SSR, CSR, UI Architecture

Web Development, UI, Debugging, DOM Manipulation

REST APIs, CRUD Systems, Third-Party APIs, Integration Workflows

Databases, Data Modeling, Query Design

Automation, Data Processing, Machine Learning, Scripting

Neural Networks, Deep Learning, Model Training, Experimentation

OOP, Basic Backend Concepts, Academic Projects

Web Applications, MVC, Backend Development

Workflow Automation, Business Process Automation, Integrations

TCP/IP, DNS, DHCP, HTTP/HTTPS, Networking Fundamentals

(Zapier,

Make,

 EDUCATION

2024 - 2026

2026 - Present

B.Sc. in Data Science and Artificial Intelligence, ESPOL, Ecuador

B.Sc. in Software Engineering (Development, Operations & Security), UEES Online, Ecuador

 ADDITIONAL INFO RMATION

Based in Ecuador. Open to remote roles or international relocation at my own expense..